

Ashton E. Thomas

EDUCATION

University of Michigan - Rackham Graduate School	08/2025 – 05/2026
<i>Major - MSE Computer Engineering — Concentration in ML/CV</i>	<i>Ann Arbor, MI, USA</i>
University of Michigan College of Engineering	08/2022 – 05/2025
<i>Major - BSE Computer Science, GPA - 3.81 / 4.0</i>	<i>Ann Arbor, MI, USA</i>
<i>Relevant coursework: AR/VR, Computer Organization, Computer Vision, Cryptography, Data Analytics</i>	
<i>Data Structures & Algorithms, Operating Systems, Quantum Computing, Web Development, Web Systems</i>	

EXPERIENCE

Amazon	09/2024 – Present
<i>Software Development Engineer</i>	<i>Boston, MA & New York City, USA</i>
- Partnered with a team of more than 20 engineers to design, test, and optimize Alexa devices, enhancing functionality and elevating user experience. - Independently developed and implemented new features for Alexa devices, optimizing performance and user experience with the use of React Native, Kotlin, TypeScript, and related technologies. - Worked in a team of 10 engineers to manage Amazon company-wide employee data regarding 100,000s employees, including the collection, processing, and inferences used by corporate executives at the top of Amazon.	
Ground Vehicle Systems Center (SEC)	05/2024 – 08/2024
<i>Software Engineer</i>	<i>Warren, MI, USA</i>
- Leveraged MagicDraw & Excel to design databases for Jira tickets and hardware, leading to improved efficiency. - Created Python scripts to parse large csvs with 1000s of datapoints to update integrated networks in Jira.	
Madi Taylor Photo	06/2021 – 07/2024
<i>Full Stack Developer</i>	<i>Traverse City, MI, USA</i>
- Developed and maintained the corporate website, crafting a cohesive user interface with HTML, CSS, and JS. - Implemented robust back-end payment solutions and form validation to streamline user transactions.	

RESEARCH

Polk Lab — University of Michigan	09/2024 – Present
<i>Computational Neuroscience Research Assistant</i>	<i>Ann Arbor, MI, USA</i>
- Collaborated with a PhD student to develop a machine learning model aimed at mimicking neural distinctiveness of the human brain, while also enabling the separation of background noise from spoken language. - Worked alongside Prof. Thad Polk to develop a script that processes CSV files with 1000s of columns, utilizing wildcard parsing to dynamically filter columns. The script includes a node hierarchy for efficient column interrelation and supports operations such as adding, removing, printing, and row filtration.	

PROJECTS

Geoguessr AI, Computer Vision	2023 – 2024
- Designed and implemented a modified ResNet-50 architecture for geographic location identification from images. - Fine-tuned the model with 61k images, addressing lighting and seasonal challenges whilst achieving ~90% accuracy.	
Google Search Engine, Web Systems	2024
- Engineered a scalable search engine leveraging a segmented inverted index implemented with MapReduce. - Integrated tf-idf for text analysis and PageRank for link analysis to improve the relevance of search results produced via a REST API.	
Instagram Clone, Web Systems	2024
- Created a server-side dynamic version of Instagram with Flask and SQL relational databases, enabling features like user authentication, content management, and interactions. - Implemented a client-side dynamic version of Instagram using JavaScript, React, and REST APIs, introducing seamless content updates, infinite scroll, and double-tap to like functionality.	
Study Group Coordinator, Quantum Computing	2024
- Designed and developed a Study Group Scheduler with Quantum algorithms leveraging Grover's algorithm for efficient group formation under CNF (Conjunctive Normal Form) constraints. - Created and implemented Bitflip and Phase Oracles to translate CNF constraints into quantum operations.	

TECHNICAL SKILLS

AJAX, ARM, ARCGIS, C, C#, C++, CSS, Excel, Flask, Git, HTML, Javascript, Jinja, Jira, JSON, Keras, Kotlin, Latex, MagicDraw, Makefile, Matlab, NumPy, Python, PyTorch, Qiskit, React/Native, SQL, Typescript, Unreal Engine, VSCode